

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

1 Define the following:

Let Ω be a non-empty set, $\mathcal{F}$ is a collection of subsets of Ω and $P : \mathcal{F} \rightarrow \mathbb{R}$ is a function.

- (1) $(\Omega, \mathcal{F}, P)$ is called a probability space if ...

- (2) A function X , defined on a probability space $(\Omega, \mathcal{F}, P)$ is called a random variable (r.v) if ...

Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$, with c.d.f F_X and law $P_X \equiv \mu_{F_X}$.

- (3) X is a *Exponential*(λ) random variable ($\lambda > 0$ is a constant) if ...

- (4) X is called a *singular* variable if ...

Let $\{X_n\}$, X be random variables defined on some $(\Omega, \mathcal{F}, P)$.

- (5) $\{X_n\}$ is a sequence of independent and identically distributed (i.i.d.) random variables if ...

- (6) X_n converges to X "almost surely (a.s)" or "with probability 1 (w.p. 1)" or "a.e (P)" if ...

- (7) X_n converges to X in probability (w.r.t P) if ...

- (8) if each $X_n \in L^1(\Omega, \mathcal{F}, P)$, then the collection $\{X_n\}$ is called uniformly integrable (UI) if ...

20

(3×8 = 24 points)

2 Let X, Y be two independent random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Recall that X and Y independent iff their joint law is the *product measure* of their marginal laws, i.e $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let f, g be two non-negative functions in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$.

- (a) If X and Y are independent and $E(X^2) < \infty$, $E(Y^2) < \infty$, then carefully prove

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

- (b) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove

$$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f \cdot g \quad \text{a.e. } (m^2)$$

— something with Tonelli

$$f_n \rightarrow f$$

- (c) If f is continuous and bounded on $\mathbb{R}$, so is $f * g$.

$$|f(x-y)| \cdot |g(y)| \leq m g(y)$$

- (d) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e. (m), f and g are as in (b).

↳ continuous / bounded

✓ (e) If Z is a random variable with $P_Z = \lambda$ as in (d) above, then how is Z related to X and Y ? What does the result in (d) say about the law of Z ?

(6+6+6+6+(3+3)= 30 points)

Z is the sum of X and Y

the law of Z is the distribution of $(X+Y)$

Let $\{X_n\}_{n \geq 1}$ be a sequence of *nonnegative* random variables defined on a probability space $(\Omega, \mathcal{F}, P)$. Let

$$Y(\omega) = \liminf_{n \rightarrow \infty} X_n(\omega), \omega \in \Omega.$$

should be

able

to

- (i) Prove $Y : \Omega \rightarrow \mathbb{R}$ is a random variable.
 (ii) If $\limsup_{n \rightarrow \infty} E(X_n) < \infty$, then prove that $Y \in L^1(\Omega, \mathcal{F}, P)$.
 (iii) If $M(t)$ is the moment generating function of Y , i.e. $M(t) = E(e^{tY})$, $t \in \mathbb{R}$, then show

$$P(Y > a) \leq \inf_{t > 0} e^{-ta} M(t), \quad a > 0.$$

Markov

(5+5+5= 15 points)

similar

to

Let m be the Lebesgue measure on $[0, 1]$ and $I_A(x)$ denote the indicator function of A . Consider the following sequence of random variables on $(\Omega = [0, 1], \mathcal{F} = \mathcal{B}[0, 1], P = m)$: For $n \geq 1$, $X_n(\omega) = nI_{(0, \frac{1}{n^2})}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.

- (i) Show $X_n \rightarrow X$ almost surely (P).
 (ii) Does $X_n \rightarrow X$ in probability?
 (iii) Show $X_n \rightarrow X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 2$, but this convergence does not hold for $2 \leq p < \infty$.
 (iv) Show that $X_n \rightarrow X$ in L^p for some $0 < p < \infty$ implies that $X_n \rightarrow X$ in L^q , for any $0 < q < p$.

Holder's

(3+3+(3+2)+5 = 16 points)

5 Let $\alpha \in \mathbb{R}, \beta > 0$ be constants.

- (i) Show that a $Normal(\alpha, \beta^2)$ random variable exists:

In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \rightarrow \mathbb{R}$, such that its c.d.f. is given by Φ where

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi\beta^2}} e^{-\frac{(u-\alpha)^2}{2\beta^2}} du, \quad x \in \mathbb{R}.$$

- (ii) Compute the following Lebesgue integral (Clearly stating any result used)

$$\int_{\Omega} X(\omega) dP(\omega).$$

Komogorov
consistency
theorem

- (iii) For each $n \geq 1$, let μ^n be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Under what conditions can we say that there is a probability measure P on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that $P(A \times \mathbb{R}^\infty) = \mu^n(A)$ for any $A \in \mathcal{B}(\mathbb{R}^n)$, for any $n \geq 1$? Describe a sequence of random variables $\{X_n : n \geq 1\}$ defined on $(\Omega = \mathbb{R}^\infty, \mathcal{F} = \mathcal{B}(\mathbb{R}^\infty), P)$ such that the joint distribution of $(X_1, \dots, X_n)$ is given by μ^n , for all $n \geq 1$.

- (iv) Using part (iii) or otherwise, show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Normal(\alpha, \beta^2)$ random variables.

(5+5+5+5= 20 points)

Let $f(\alpha_i) = \Phi(x)$
or $\mu = \Phi(x)$

1) Ω a non-empty set, $\mathcal{A}$ a collection of subsets of Ω , and $P: \mathcal{A} \rightarrow \mathbb{R}$ is a function

(1) $(\Omega, \mathcal{A}, P)$ is called a probability space if $\mathcal{A}$ is a σ -algebra and P is a probability measure s.t. $P(\Omega) = 1$.

(2) A function X defined on a probability space $(\Omega, \mathcal{A}, P)$ is called a random variable if $X: \Omega \rightarrow \mathbb{R}$ and X is $(\mathcal{A}, \mathcal{B}(\mathbb{R}))$ -measurable. That is

$$X^{-1}((-\infty, a]) \in \mathcal{A} \quad \text{for } a \in \mathbb{R}.$$

(3) X is an exponential(λ) random variable ($\lambda > 0$ a constant) if

$$P_X = \mu_F \quad \text{where } F \text{ is the CDF of an exponential } (\lambda) \text{ r.v.}$$

(4) X is a singular random variable if P_X is singular wrt m . That is if $\exists$ some set $B \in \mathcal{B}(\mathbb{R})$ s.t. $P_X(B) = 0 \Rightarrow m(B^c) = 0$.

(5) $\{X_n\}$ is a sequence of iid random variables if

$$P_{(X_1, \dots, X_n)} = P_{X_1} \times \dots \times P_{X_n} \quad \forall n \geq 1 \text{ and } P_{X_i} = P_{X_j} \text{ for all } i \neq j.$$

(6) X_n converges to X almost surely if $\exists$ some $B \in \mathcal{B}(\mathbb{R})$ s.t. $P(B) = 0$ and

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$$

for all $\omega \in B^c$.

(7) X_n converges to X in probability if $\forall \varepsilon > 0$

$$\lim_{n \rightarrow \infty} P(\{\omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) = 0.$$

(8) If each $X_n \in L^1(\Omega, \mathcal{F}, P)$, then the collection $\{X_n\}$ is called uniformly integrable if

$$\sup_{n \in \mathbb{N}} a_n(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

where

$$a_n(t) = \int_{\{|X_n| > t\}} |X_n| dP.$$

2) X, Y independent r.v.s on $(\Omega, \mathcal{F}, P)$ w/ marginal laws $P_X + P_Y$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

Q. X, Y indep. $E[X^2] < \infty, E[Y^2] < \infty \rightarrow$ Prove

$$\text{Var}[X+Y] = \text{Var}[X] + \text{Var}[Y]$$

By definition

$$\begin{aligned} \text{Var}[X+Y] &= E[(X+Y) - (E[X+Y])]^2 \\ &= E[(X+Y) - (E[X] + E[Y])]^2 \end{aligned}$$

$$= E[(X+Y - E[X] - E[Y])^2]$$

$$= E[(X - E[X] + Y - E[Y])^2]$$

$$= E[(X - E[X])^2 + (Y - E[Y])^2 + 2(X - E[X])(Y - E[Y])]$$

$$= \int [(X - E[X])^2 + (Y - E[Y])^2 + 2(X - E[X])(Y - E[Y])] dP_{X,Y}$$

$$= \int (X - E[X])^2 dP_{X,Y} + \int (Y - E[Y])^2 dP_{X,Y} + 2 \int (X - E[X])(Y - E[Y]) dP_{X,Y}$$

$$= \iint (X - E[X])^2 dP_X dP_Y + \iint (Y - E[Y])^2 dP_X dP_Y$$

$$+ 2 \int (XY - XE[Y] - E[X]Y + E[X]E[Y]) dP_{X,Y}$$

$$= \int (X - E[X])^2 dP_X + \int (Y - E[Y])^2 dP_Y$$

$$+ 2 \left[\int XY dP_{X,Y} - E[Y] \int X dP_{X,Y} - E[X] \int Y dP_{X,Y} + \int E[X]E[Y] dP_{X,Y} \right]$$

$$= \text{Var}[X] + \text{Var}[Y] + 2[E[XY] - E[Y]E[X] - E[X]E[Y] + E[X]E[Y]]$$

$$= \text{Var}[X] + \text{Var}[Y] + 2[E[XY] - E[Y]E[X]] \rightarrow$$

$$\begin{aligned} E[X+Y] &= \int X+Y dP_{X,Y} \\ &= \int X dP_X + \int Y dP_Y \end{aligned}$$

$$\begin{aligned} &= \iint X dP_X dP_Y + \iint Y dP_X dP_Y \\ &\quad \begin{array}{l} \text{by Fubini's} \\ \text{since } \int X dP_X < \infty \\ \text{since } \int Y dP_Y < \infty \\ \text{+ same for } Y \end{array} \\ &= \int X dP_X + \int Y dP_Y \\ &= E[X] + E[Y] \end{aligned}$$

Now consider $E[XY]$.

$$E[XY] = \int XY dP_{X,Y}$$

$$= \int XY \cdot dP_X \times P_Y \quad \text{by independence of } X \text{ and } Y$$

$$= \iint XY dP_X dP_Y \quad \text{by Fubini's since}$$

$$= \int X dP_X \int Y dP_Y$$

$$= E[X] E[Y]$$

$$\int |XY| dP_{X,Y} \leq \left(\int |X|^2 dP_{X,Y} \right)^{1/2} \left(\int |Y|^2 dP_{X,Y} \right)^{1/2}$$

Cauchy-Schwarz inequality $< \infty$ since $E[X^2] < \infty$ and $E[Y^2] < \infty$.

Thus,

$$\begin{aligned} \text{Var}[X+Y] &= \text{Var}[X] + \text{Var}[Y] + 2(E[X]E[Y] - E[XY]) \\ &= \text{Var}[X] + \text{Var}[Y]. \end{aligned}$$

(b) $P_X \ll m$ and $P_Y \ll m$ w/ $\frac{dP_X}{dm} = f$ + $\frac{dP_Y}{dm} = g$.

Want to prove $P_{(X,Y)} \ll m^2$ and $\frac{dP_{(X,Y)}}{dm^2} = f \cdot g$ a.e. (m^2)

$P_X \ll m$

$$\Rightarrow P_X(A_1) = \int_{A_1} f(x) dm(x) \quad \forall A_1 \in \mathcal{B}(\mathbb{R})$$

$P_Y \ll m$

$$\Rightarrow P_Y(A_2) = \int_{A_2} g(x) dm(x) \quad \forall A_2 \in \mathcal{B}(\mathbb{R})$$

Then for $A = A_1 \times A_2 \in \mathcal{B}(\mathbb{R}^2)$,

$$P_{X,Y}(A) = P_X(A_1) \times P_Y(A_2) \quad \text{by independence}$$

$$= \int_{A_1} f(x) dm(x) \int_{A_2} g(y) dm(y)$$

$$= \int_{A_2} \int_{A_1} f(x)g(y)dm(x)dm(y)$$

$$= \int_{A_1 \times A_2} f(x)g(y)d(m(x) \times m(y)) \quad \text{by Fubini's since } f, g \in L^1(\mathbb{R}, (\mathcal{B}(\mathbb{R}), m))$$

$$= \int_{A_1 \times A_2} f(x)g(y)d(m^2(x, y))$$

Consider that since $f + g$ are both non-negative functions, $f \cdot g$ is also a non-negative function. Thus, since we are able to write $P_{X,Y}$ in this manner, it follows that

$$P_{X,Y} \ll m^2,$$

and by the Radon-Nikodym derivative

$$\frac{dP_{X,Y}}{dm^2} = f \cdot g$$

a.e. (m^2) by the uniqueness of the R-N derivative.

(c) f is continuous and bounded on $\mathbb{R}$. WTS that $f * g$ is also continuous and bounded on $\mathbb{R}$

Let us start by showing that $f * g$ is bounded. Consider that f is bounded, so let $M \in \mathbb{R}$, $M > 0$ s.t. $|f(x)| \leq M$.

$\forall x \in \mathbb{R}$. Then

$$\begin{aligned} |f * g(x)| &= \left| \int f(x-u)g(u)dm(u) \right| \\ &\leq \int |f(x-u)g(u)|dm(u) \\ &\leq \int |Mg(u)|dm(u) \end{aligned}$$

$$= M \int |g(u)| dm(u)$$

$$< \infty$$

Since $g \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$. Thus, $f * g$ is bounded by

$$M \cdot \int |g| dm$$

Now let us show that $f * g$ is continuous. Let $\{x_n\}_{n \geq 1}$ be a sequence s.t. $x_n \rightarrow x$ as $n \rightarrow \infty$. Now consider the function $h(x) = (x - u)$. This is a continuous function since $h(x_n) = x_n - u \rightarrow x - u$ as $n \rightarrow \infty$. Then we are told that f is continuous, so

$$f(h(x_n)) = f(x_n - u) \rightarrow f(x - u) \text{ as } n \rightarrow \infty.$$

Finally, we can think of $g(u)$ as a constant, so

$$f(x_n - u)g(u) \rightarrow f(x - u)g(u) \text{ as } n \rightarrow \infty.$$

We can now apply the DCT. That is since

$$(i) |f(x_n - u)g(u)| \leq M |g(u)| \quad (|f(x_n - u)g(u)| \text{ is bounded for all } x_n \in \mathbb{R} \text{ and } n \geq 1,$$

$$(ii) g(u) \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m), \text{ and}$$

$$(iii) f(x_n - u)g(u) \rightarrow f(x - u)g(u) \text{ as } n \rightarrow \infty,$$

we can apply the DCT to say that

$$\int f(x_n - u)g(u) dm(u) \rightarrow \int f(x - u)g(u) dm(u)$$

$$\text{where } f * g = \int f(x - u)g(u) dm(u). \text{ Thus } f * g(x_n) \rightarrow f * g(x).$$

f is continuous!

f * g is continuous!

(d) If $\lambda = P_X * P_Y$, we want to prove that $\frac{d\lambda}{dm} = f * g$ a.e. (m).

$$\lambda(A) = P_X * P_Y(A)$$

$$= \iint I_A(x+y) dP_X(x) dP_Y(y)$$

$$= \iint I_A(x+y) f(x) dm(x) dP_Y(y)$$

$$\frac{dP_X}{dm} = f$$

$$\Rightarrow dP_X = f \cdot dm$$

$$= \iint I_A(u) f(u-y) dm(u) dP_Y(y)$$

$$x+y=u$$

$$\Rightarrow x = u-y$$

$$= \iint I_A(u) f(u-y) dP_Y(y) dm(u)$$

by Tonelli since

f & I_A are non-neg. functions

$$= \int I_A(u) \left[\int f(u-y) dP_Y(y) \right] dm(u)$$

$$= \int_A \int f(u-y) g(y) dm(y) dm(u)$$

$$\frac{dP_Y}{dm} = g$$

$$\Rightarrow dP_Y = g dm$$

$$= \int_A [f * g(u)] dm(u).$$

Now since we can write λ in this form where $f * g$ is a non-negative function, we get that

$$d\lambda \ll dm$$

and by the Radon-Nikodym derivative

$$\frac{d\lambda}{dm} = f * g$$

a.e. (m) by the uniqueness of the Radon-Nikodym derivative.

(e) Z a random variable with $P_Z = \lambda$. $\rightarrow$ because of the previous problem, we have that

$$Z = X + Y$$

and Z has a probability distribution^(law) of

$$P_Z = f * g.$$

(which is the distribution of $X + Y$).

③ $\{X_n\}_{n \geq 1}$ is a sequence of non-negative r.v.s defined on a probability space $(\Omega, \mathcal{F}, P)$. Let

$$Y(\omega) = \liminf_{n \rightarrow \infty} X_n(\omega), \quad \omega \in \Omega.$$

(i) WTS that $Y: \Omega \rightarrow \mathbb{R}$ is a random variable.

That is, we need to show that $Y(\omega)$ is $(\mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}))$ -measurable. We will start by showing that

$\inf_{n \geq 1} X_n(\omega)$ is $(\mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}))$ -measurable.

Let $g = \sup_{n \geq 1} X_n(\omega)$. Then

$$\begin{aligned} g^{-1}((-\infty, r]) &= \{\omega : g(\omega) \leq r\} \\ &= \{\omega : \sup_{n \geq 1} X_n(\omega) \leq r\} \\ &= \{\omega : X_n(\omega) \leq r \quad \forall n \geq 1\} \\ &= \bigcap_{n \geq 1} \{\omega : X_n(\omega) \leq r\} \\ &= \bigcap_{n \geq 1} X_n^{-1}((-\infty, r]) \end{aligned}$$

We know that X_n is $(\mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}))$ -meas since it is a random variable for all $n \geq 1$. Thus,

$X_n^{-1}((-\infty, r]) \in \mathcal{F}$ $\forall n \geq 1$. Then since $\mathcal{F}$ is a σ -algebra

$$\bigcap_{n \geq 1} X_n^{-1}((-\infty, r]) \in \mathcal{F}$$

since σ -algebras are closed under countable intersections. Thus, $\sup_{n \geq 1} X_n(\omega)$ is $(\mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}))$ -meas. $\rightarrow$

Consider that

$$\inf_{n \geq 1} X_n(\omega) = - \sup_{n \geq 1} (-X_n(\omega)).$$

Thus, $\inf_{n \geq 1} X_n(\omega)$ is a composition of meas. functions,
so $\inf X_n$ is also $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable.

Furthermore, consider that

$$\liminf_{n \rightarrow \infty} X_n(\omega) = \sup_{n \geq 1} \left(\inf_{n \geq j} X_n(\omega) \right).$$

Thus, $\liminf_{n \rightarrow \infty} X_n$ is also a composition of
measurable functions, so $\liminf_{n \rightarrow \infty} X_n(\omega)$ is

$(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable.

(ii) If $\limsup_{n \rightarrow \infty} E[X_n] < \infty$, prove that $\forall Y \in L^1(\Omega, \mathcal{F}, P)$. That
is, we want to show that $\int |Y| dP < \infty$. Consider
the following...

$$\int |Y| dP = \int \left| \liminf_{n \rightarrow \infty} X_n(\omega) \right| dP$$

$$= \int \liminf_{n \rightarrow \infty} X_n(\omega) dP \quad (\text{since } X_n \text{'s are non-negative})$$

$$\leq \liminf_{n \rightarrow \infty} \int X_n(\omega) dP \quad (\text{by Fatou's lemma since } \{X_n\}_{n \geq 1} \text{ is a seq. of non-neg. meas. functions})$$

$$\leq \limsup_{n \rightarrow \infty} \int X_n(\omega) dP \\ = \limsup_{n \rightarrow \infty} E[X_n] < \infty \quad \checkmark$$

(iii) Let $M(t)$ be the moment generating function of Y .

$$(M(t) = E[e^{tY}], t \in \mathbb{R}). \text{ WTS}$$

$$P(Y > a) \leq \inf_{t \geq 0} e^{-ta} M(t), \quad a > 0.$$

Consider that Cramer's Inequality says that for X a random variable and $t > 0$,

$$P(X \geq t) \leq \inf_{\theta > 0} \frac{E[e^{\theta X}]}{e^{\theta t}}.$$

Since Y is a random variable (as proved in (i)) and $a > 0$, we can apply Cramer's inequality.

$$P(Y > a) \leq \inf_{t \geq 0} \frac{E[e^{tY}]}{e^{ta}}$$

$$= \inf_{t \geq 0} e^{-ta} M(t).$$

✓

④ $([0, 1], \mathcal{B}([0, 1]), m)$ $X_n(\omega) = n \mathbb{I}_{(0, \frac{1}{n^2})}(\omega)$ for $n \geq 1$ & $X(\omega) = 0$ for $\omega \in [0, 1]$.

(i) WTS $X_n \rightarrow X$ a.s. (P). That is, we want to find some $B \in \mathcal{B}([0, 1])$ s.t. $m(B) = 0$ and $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \forall \omega \in B^c$. Consider that

$$X_n(\omega) = \begin{cases} 0 & \text{when } \omega = 0 \\ n & \text{when } \omega \in (0, \frac{1}{n^2}) \\ 0 & \text{when } \omega \in [\frac{1}{n^2}, 1]. \end{cases}$$

Thus,

$$X_n(0) = 0 \quad \forall n \geq 1,$$

$$X_n(1) = 0 \quad \forall n \geq 1, \text{ and}$$

for $\omega \in (0, 1)$, $\exists$ some $N \in \mathbb{N}$ s.t. $\frac{1}{N^2} \leq \omega$ so that

$$X_n(\omega) = 0 \quad \forall n \geq N.$$

Thus, for $B = \emptyset$ and $m(\emptyset) = 0$,

$$\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) = 0 \quad \forall \omega \in B^c = [0, 1].$$

Thus, $X_n \rightarrow X$ a.s. (P).

(ii) Now, we want to determine if $X_n \rightarrow X$ in probability. That is, we want to show that $\forall \varepsilon > 0$

$$\lim_{n \rightarrow \infty} m(\{\omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) = 0.$$

Consider that ...



$$\lim_{n \rightarrow \infty} m(\{\omega: |X_n(\omega) - X(\omega)| > \varepsilon\})$$

$$= \lim_{n \rightarrow \infty} m(\{\omega: |X_n - 0| > \varepsilon\})$$

$$= \lim_{n \rightarrow \infty} m(\{\omega: X_n(\omega) > \varepsilon\})$$

$$= \lim_{n \rightarrow \infty} m(\{\omega: \omega \in (0, \frac{1}{n^2})\})$$

$$= \lim_{n \rightarrow \infty} m(0, \frac{1}{n^2})$$

$$= \lim_{n \rightarrow \infty} (\frac{1}{n^2} - 0)$$

since m is the Lebesgue measure.

$$= \lim_{n \rightarrow \infty} \frac{1}{n^2}$$

$$= 0$$

Thus, $X_n \rightarrow X$ in probability. ✓

(iii) WTS $X_n \rightarrow X$ in $L^p(\Omega, \mathcal{F}, P)$ for $p=1$ ($0 < p < 2$)

That is, we need to show that

$$(1) \int |X_n|^1 dm < \infty \quad \forall n \geq 1,$$

$$(2) \int |X|^1 dm < \infty, \text{ and}$$

$$(3) \lim_{n \rightarrow \infty} \int |X_n(\omega) - X(\omega)|^1 dm = 0.$$

Consider that

$$\int |X_n| dm = \int n \cdot \mathbb{I}_{(0, \frac{1}{n^2})} dm$$

$$= n \cdot m((0, \frac{1}{n^2})) = n \cdot (\frac{1}{n^2}) = \frac{1}{n} < \infty \quad \forall n \geq 1.$$

We also have that

$$\int |X|' dm = \int 0' dm = 0 < \infty. \quad \checkmark$$

Finally,

$$\lim_{n \rightarrow \infty} \int |X_n - X| dm = \lim_{n \rightarrow \infty} \int |X_n| dm = \lim_{n \rightarrow \infty} \frac{1}{n} = 0. \quad \checkmark$$

Hence, $X_n \rightarrow X$ in L^1 .

However, we also want to show that $X_n \not\rightarrow X$ in L^p
for $2 \leq p < \infty$

$$\begin{aligned} \int |X_n|^p dm &= \int |n \cdot I_{(0, \frac{1}{n^2})}|^p dm \\ &= \int n^p \cdot I_{(0, \frac{1}{n^2})} dm \\ &= n^p m\left(0, \frac{1}{n^2}\right) \\ &= \frac{n^p}{n^2} \\ &= n^{p-2} < \infty \quad \forall n \geq 1. \end{aligned}$$

$$\int |X|^p dm = \int 0 dm = 0$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \int |X_n - X|^p dm &= \lim_{n \rightarrow \infty} \int |X_n|^p dm \\ &= \lim_{n \rightarrow \infty} n^{p-2} \end{aligned}$$

$$> 0 \quad \text{when} \quad 2 \leq p < \infty.$$

Hence, $X_n \not\rightarrow X$ in L^p for $2 \leq p < \infty$.

(iv) WTS if $X_n \rightarrow X$ in L^p for some $0 < p < \infty$, then $X_n \rightarrow X$ in L^q for any $0 < q < p$.

Let $Y = |X_n - X|^q$ and $Z = 1$. Further, let $r = \frac{p}{q}$ and $s = \frac{r}{1-r}$.
Let us first show that $Y \in L^r$ and $Z \in L^s$, so that we can apply Hölder's inequality.

$$\begin{aligned} \int |Y|^r dm &= \int ||X_n - X|^q|^r dm \\ &= \int |X_n - X|^{q \cdot (\frac{p}{q})} dm \\ &= \int |X_n - X|^p dm \\ &< \infty \end{aligned}$$

$$\begin{aligned} \int |Z|^s dm &= \int |1|^s dm \\ &= m(0,1) \\ &= 1 - 0 \\ &= 1 \\ &< \infty \end{aligned}$$

By Hölder's inequality,

$$\begin{aligned} \int |Y \cdot Z| dm &\leq \left(\int |Y|^r dm \right)^{\frac{1}{r}} \left(\int |Z|^s dm \right)^{\frac{1}{s}} \\ \Rightarrow \int |Y \cdot 1| dm &\leq \left(\int ||X_n - X|^q|^r dm \right)^{\frac{1}{r}} \left(\int |1|^s dm \right)^{\frac{1}{s}} \\ \Rightarrow \int |X_n - X|^q dm &\leq \left(\int |X_n - X|^p dm \right)^{\frac{1}{r}} (1) \\ \Rightarrow \lim_{n \rightarrow \infty} \int |X_n - X|^q dm &\leq \lim_{n \rightarrow \infty} \left(\int |X_n - X|^p dm \right)^{\frac{1}{r}} \\ &= 0 \Rightarrow \lim_{n \rightarrow \infty} \int |X_n - X|^q dm = 0 \end{aligned}$$

for any $0 < q < p$.
 $X_n \rightarrow X$ in L^q

⑤ (i) WTS that a $N(0,1)$ random variable exists...

Let $F(x) = \Phi(x)$ where Φ is the CDF of a $N(0,1)$ r.v., which is a non-decreasing, right-continuous function. Then let

$$\mathcal{C} = \{(a, b] : -\infty \leq a \leq b \leq \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\}.$$

and define

$$\mu_F((a, b]) = \mu_\Phi((a, b]) = \Phi(b+) - \Phi(a+) \quad \text{and}$$

$$\mu_F((a, \infty)) = \mu_\Phi((a, \infty)) = \Phi(\infty) - \Phi(a+).$$

It can be shown that

(1) $\mathcal{C}$ is a semi-algebra

(2) μ_Φ is a measure on $\mathcal{C}$.

Now define μ_Φ^* as

$$\mu_\Phi^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu_\Phi(A_n) : \{A_n\}_{n=1}^{\infty} \subset \mathcal{C}, A \subset \bigcup_{n=1}^{\infty} A_n \right\}.$$

Then by Caratheodory's extension theorem:

(1) μ_Φ^* is an outer measure

(2) $\mathcal{C} \subset \mathcal{M}_{\mu_\Phi^*}$ where $\mathcal{M}_{\mu_\Phi^*} = \{A : A \text{ is } \mu^* \text{-measurable}\}$ and

by another result:

(i) $\mathcal{M}_{\mu_\Phi^*}$ is a σ -algebra

(ii) μ_Φ^* restricted to $\mathcal{M}_{\mu_\Phi^*}$ is a measure

(iii) $\mu^*(A) = 0 \Rightarrow P(A) \in \mathcal{M}_{\mu_\Phi^*}$ (says that $(\mathbb{R}, \mathcal{M}_{\mu_\Phi^*}, \mu_\Phi^*)$ is a complete measure)

(3) $\mu_\Phi^* = \mu_\Phi$ on $\mathcal{C}$.

Thus, we now have a measure space $(\mathbb{R}, \mathcal{M}_{\mu_\Phi^*}, \mu_\Phi^*) \rightarrow$

Consider that $e \in \mathcal{M}_{\mathbb{R}}^* \Rightarrow \sigma(e) \in \mathcal{M}_{\mathbb{R}}^*$ and $\sigma(e) = \mathcal{B}(\mathbb{R})$. Thus, $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_{\mathbb{R}}^*)$ is also a measure space, and $\mu_{\mathbb{R}}^* = \mu_{\mathbb{R}}$ on $\mathcal{B}(\mathbb{R})$. Note that $\mu_{\mathbb{R}}^*$ is unique by the uniqueness of measures theorem. Now let $X(\omega) = \omega \ \forall \ \omega \in \mathbb{R}$. Consider that $X: \mathbb{R} \rightarrow \mathbb{R}$ and is a meas. function since it is a constant. Further,

$$\mu_{\mathbb{R}}^*(\mathbb{R}) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 1,$$

so X is a random variable with

$$F_X = \Phi \quad \text{and} \quad P_X = P(X^{-1}(\cdot)) = \mu_{\mathbb{R}}^*, \\ = \mu_{\mathbb{R}}^*((-\infty, a])$$

Thus, a $N(0,1)$ random variable exists.

(ii) Want to compute $\int_{\Omega} X(\omega) dP(\omega)$

$$\int_{\Omega} X(\omega) dP(\omega) = \int_{\Omega} I(X(\omega)) dP(\omega) \quad I(x) = x$$

$$= \int_{\mathbb{R}} I(\omega) dP \circ X^{-1}(\omega) \quad * \text{ change of variables}$$

$$= \int_{\mathbb{R}} \omega dP_X(\omega)$$

$$\frac{dP_X}{d\omega} = \delta$$

$$\Rightarrow dP_X = \delta d\omega$$

$$= \int_{\mathbb{R}} \omega \cdot \delta(\omega) d\omega$$

* R-N derivative

$$= \int_{-\infty}^{\infty} \omega \cdot \frac{1}{\sqrt{2\pi}} e^{-\omega^2/2} d\omega = 1$$

(iii) There will be a probability meas on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ s.t. $P(A \times \mathbb{R}^\infty) = \mu_n(A) \quad \forall A \in \mathcal{B}(\mathbb{R}^n)$ for $n \geq 1$ if

$$\mu_{n+1}(A \times \mathbb{R}) = \mu_n(A) \quad \forall A \in \mathcal{B}(\mathbb{R}^n) \quad \forall n \geq 1.$$

(By Kolmogorov's consistency theorem).

Then let $X_n(\omega)$ for $\omega = (\omega_1, \omega_2, \dots) \in \mathbb{R}^\infty$ be defined as $X_n(\omega) = \omega_n$.

Then

$$P_{(X_1, \dots, X_n)}(A) = \mu_n(A) \quad \forall A \in \mathcal{B}(\mathbb{R}^n), n \geq 1.$$

(iv) WTS $\exists$ a sequence $\{X_n: n \geq 1\}$ of iid $N(0,1)$ r.v.s. Let $F = \Phi$ be the normal $(0,1)$ CDF and $P_X = \mu_F$ be the probability distribution of $X \sim N(0,1)$. Then define

$$\mu_1 = \mu_F$$

$$\mu_2 = \mu_F \times \mu_F$$

$\vdots$

$$\mu_n = \underbrace{\mu_n \times \mu_F}_n = \mu_F \times \dots \times \mu_F$$

Note that the consistency conditions hold:

$$(1) \mu_n(\mathbb{R}^n) = \underbrace{\mu_F(\mathbb{R}) \times \dots \times \mu_F(\mathbb{R})}_n = 1 \times \dots \times 1 = 1$$

so μ_n is a probability measure

$$(2) \mu_{n+1}(B \times \mathbb{R}) = \mu_n(B) \times \mu_F(\mathbb{R}) = \mu_n(B) \times 1 = \mu_n(B) \quad \forall B \in \mathcal{B}(\mathbb{R}^n) \text{ and } n \geq 1.$$

Then K's cons. thm tells us that there exists

probability measure P on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ s.t. for all $n \geq 1$ $P_{(X_1, \dots, X_n)} = \mu_n$. Thus, $\{X_n\}$ is a seq. of iid r.v.s.